

SnipKeep

Ten features to make SnipKeep life-changing for learners

A research pass on real student problems, and how AI — used with discipline — turns a clipping tool into a learning system. Written from four seats at once: innovator, UX designer, psychological scientist, learning coach.

Internal product research · July 2026 · extension + proposed web companion

The spine: why SnipKeep can win where AI study tools fail

The uncomfortable research fact: in the landmark Dunlosky review of ten study techniques, the two things students do most — highlighting and rereading — rank **lowest** in effectiveness. The two that rank highest are **retrieval practice** (recalling from memory) and **spaced practice** (returning over days). SnipKeep is, literally, a highlighter. That is either its weakness or its superpower.

The 2026-specific trap: students now ask AI to explain things, feel instant clarity, and move on — the *illusion of understanding*. Clarity in the moment is not durable knowledge; the feeling of knowing is exactly what retrieval practice exists to test. Most AI study tools amplify this trap. SnipKeep's existing ground rule — **AI asks and classifies; it never digests for the student** — is precisely the antidote, and no major tool is built on it.

The unfair advantage: SnipKeep owns the exact moment a learner marks something as important. Every feature below uses that moment to quietly convert the worst study habit (collecting) into the best ones (retrieving, spacing, explaining, connecting) — without ever feeling like homework.

Rule 1 — AI asks, never answers-for-you.

Every AI feature outputs questions, classifications, or gaps. If a feature would write the summary, analogy, or explanation on the student's behalf, it is producing passive consumption and doesn't ship.

Rule 2 — Ownership stays absolute.

The extension stays serverless. The web companion is local-first: data lives in the user's own Google Doc/Drive, AI runs on their own key. Export is always one click. Investment must never become lock-in.

Rule 3 — No shame mechanics.

No streaks, no guilt, no red badges. Spacing and review are invitations with real psychology behind them, not compliance systems.

The ten features

01 Retrieval Flip — every highlight becomes a question

EXTENSION

PROBLEM	Highlighting feels productive but barely helps memory; students re-see clips and re-read them, the lowest-utility loop in the literature.
FEATURE	At save time, AI silently drafts one retrieval question whose answer is the clip ("What were the two reasons Chrome's PDF viewer blocks extensions?"). Anywhere the clip would be shown again — history, resurfacing — the question shows <i>first</i> ; the clip reveals on tap. Saving is unchanged; the flip happens on re-encounter.
SCIENCE	Practice testing is the #1-rated technique in Dunlosky's review; even one forced recall attempt beats multiple rereads.
WHY US	Only a tool that owns the highlight moment can do this invisibly, with zero card-writing friction — the reason Anki adoption fails.

02 The Five-Minute Review — a daily spaced-retrieval inbox

WEB

PROBLEM	Notes are write-only memory: saved once, never revisited. Spacing is the second-strongest technique and almost nobody self-administers it.
FEATURE	The web companion's home screen is just today's 3–5 due questions (from Feature 01), self-graded "got it / not yet," scheduled by a spacing algorithm running entirely client-side. Five minutes, then it tells you you're done. Done means done — no infinite feed.
SCIENCE	Distributed retrieval ("successive relearning") shows some of the largest effects in real classrooms. Self-grading keeps the generation step honest.
WHY US	The queue builds itself from reading the student already did. Cold-start problem — the death of every flashcard app — doesn't exist.

03 Teach-It-Back — Feynman mode on the existing voice engine

WEB + EXTENSION

PROBLEM	Students can't tell the difference between recognizing material and being able to produce it — until the exam does it for them.
FEATURE	Pick a Doc, hit the mic (infrastructure already built for voice notes), and explain the topic out loud from memory . AI transcribes, maps the explanation against the actual clips, and returns only: covered / missing / possibly-misremembered — each gap phrased as a question ("You didn't mention why the control group mattered — why did it?"). It never supplies the missing explanation.
SCIENCE	Self-explanation is a high-utility technique; production tests reveal fluency illusions that recognition can't.
WHY US	The comparison target is the student's own curated clips — not generic textbook content. No other tool has that corpus.

04 Predict-First — prediction prompts on lecture video

EXTENSION

PROBLEM	Video lectures are consumed passively at 2× speed; retention after a week is a fraction of what students believe.
FEATURE	On watch pages (timestamp-clipping already exists), an opt-in mode drops one prediction prompt at a natural chapter boundary: "Before continuing — what do you think happens to the interest rate here, and why?" Typed or spoken in five seconds, never graded, kept with the timestamp. The video then answers it.
SCIENCE	The prediction effect: generating a guess before learning — even a wrong one — measurably strengthens encoding of the correction.
WHY US	SnipKeep already lives inside the video page with timestamps; prompts anchor to real moments, and wrong predictions become the best Feature-01 questions.

05 The Confusion Flag — "I don't get this yet"

EXTENSION + WEB

PROBLEM	Confusion is the highest-value signal in learning and every tool ignores it; students save what they don't understand and quietly never return.
FEATURE	One tap marks a clip "don't get it yet." AI responds with a prerequisite ladder — two or three questions ordered from what the student demonstrably knows (their own archive) up to the confusing clip — plus, when possible, "you saved something in [other Doc] that works as an analogy — try mapping them." The student climbs; the ladder never explains.
SCIENCE	Scaffolding within the zone of proximal development; analogical encoding is strongest when the learner constructs the mapping themselves.
WHY US	Personalized prerequisite paths need to know what this student has already read. SnipKeep knows. A chatbot doesn't.

06 Contradiction Radar

EXTENSION

PROBLEM	Students collect sources that disagree without noticing, then write essays with incoherent claims — source evaluation is the least-taught research skill.
FEATURE	When a new clip makes a claim that conflicts with an existing one, a quiet save-time flag: "This appears to disagree with something you saved in March." Both clips side by side, one question: "Which do you trust more — and what would settle it?" The answer is saved as a margin note. AI detects the conflict; the student adjudicates it.
SCIENCE	Epistemic cognition training; elaborative interrogation ("why is this true?") is a top-tier technique, and genuine contradictions are its most natural trigger.
WHY US	Claim-level comparison across a personal archive is uniquely possible here — and it makes the archive smarter as it grows, deepening the ownership moat.

07 Argument Skeleton — build the essay's bones from your clips

WEB

PROBLEM	The gap between "pile of research" and "first draft" is where essays die; AI-written drafts teach nothing and get flagged.
FEATURE	A canvas where the Doc's clips appear as cards, AI-labeled by role only — claim / evidence / counterpoint / definition / procedure (the PACER insight, applied correctly). The student drags them into an argument tree; unplaced evidence and unsupported claims are visibly orphaned. Export the skeleton to the Doc and write from it — in their own words.
SCIENCE	Relational organization and dual coding; the generation effect protects the actual writing. AI classifies, the learner structures.
WHY US	This is the anti-ghostwriter: visible thinking students can defend in class, built from sources they actually read and can cite (citations already automated).

08 Exam Forge — a practice test from your own reading

WEB

PROBLEM	Students prep by rereading notes; practice tests are proven far superior but writing your own exam is skilled, slow work.
FEATURE	Any Doc with a deadline (deadline system already shipped) offers " Forge a practice exam ": mixed retrieval, application, and why-questions generated only from that Doc's clips and notes. After the attempt, AI does not grade — it classifies each answer against the source clips: covered / missed / conflicts-with-what-you-saved, with links back. The judgment stays with the student.
SCIENCE	Practice testing with feedback, aligned to the tested material; classification-not-scoring avoids outsourcing metacognition.
WHY US	Deadline + archive + citations already exist; the exam writes itself from the student's own semester. Generic quiz apps can't know what "their semester" is.

09 Knowledge Heat — the honesty dashboard

WEB

PROBLEM	The illusion of knowing: students feel prepared because they've <i>collected</i> a lot. Feeling and evidence diverge, and nothing shows them the gap.
FEATURE	Per topic, two quiet bars: collected (clips saved) vs retrieved (Feature-01/02 questions actually answered). The divergence zones — heavily saved, never successfully recalled — are named plainly: "You've saved 14 clips on monetary policy and recalled 1. This topic feels more known than it is." Retrospective, no streaks, no red.
SCIENCE	Metacognitive calibration: judgments of learning are reliably overconfident; objective retrieval data is the only known corrective.
WHY US	Requires both the collection log and the retrieval log — which only exist together in this system. It also quantifies SnipKeep's own value honestly.

10 Study Pact — deadlines become if-then plans

EXTENSION + WEB

PROBLEM	Students don't fail from ignorance of <i>what</i> to study but from never deciding <i>when</i> ; deadlines produce anxiety, then cramming — the anti-spacing.
FEATURE	When a deadline is set, one optional question: "When do you actually study? (e.g., Tue/Thu after lunch)." SnipKeep distributes the Doc's review load across those slots — "Tuesday: 10 minutes, Feature-02 questions on chapters 3–4" — and exports them as a plain calendar file (.ics, no server). Slots are invitations; a missed one just redistributes, silently.
SCIENCE	Implementation intentions ("if it's Tuesday after lunch, then I review X") roughly double follow-through on goals; distribution operationalizes spaced practice.
WHY US	It composes three things already built — deadlines, the archive, and (with Feature 02) a review queue — into the thing students actually lack: a default plan.

How to read this list

BUILD ORDER	FEATURE	REASON
1st	01 · Retrieval Flip	Smallest build, biggest philosophical shift; features 02, 08, 09 all feed on it.
2nd	02 · Five-Minute Review	The web companion's reason to exist, and the daily-return loop.
3rd	03 · Teach-It-Back	Reuses shipped voice infrastructure; the most demo-able "wow" of the ten.
Later	04–10	Each stands alone; ship one at a time, validate with real students before the next.

What was deliberately left out: anything where AI writes summaries, explanations, analogies, or drafts for the student (produces the passive consumption these features exist to fight); social/leaderboard mechanics (shame); and any feature requiring a SnipKeep-owned backend (the trust thesis is the moat — the web companion stays local-first on the user's own Drive and AI key).

The one-line pitch that falls out of this: every study tool promises to make learning easier. SnipKeep makes it *harder in exactly the ways that work* — and invisible everywhere else.

Key sources: Dunlosky et al., "Improving Students' Learning with Effective Learning Techniques" (Psych. Science in the Public Interest, 2013) · APA, "Practice Tests, Spaced Practice, and Successive Relearning" · 2026 reporting on AI study tools and the illusion-of-understanding failure mode · SnipKeep internal PACER critique (docs/IDEAS.md).